Local–Global Cross Fusion Network With Gaussian-Initialized Learnable Positional Prompting for Hyperspectral Image Classification

Xin Zhang, Rui Zhang, Ling Li, *Senior Member, IEEE*, and Wei Li

***Abstract*— Deep learning has significantly advanced the field of hyperspectral remote sensing image classification. Among various methods, the classification method based on spectral–spatial features for hyperspectral classification has attracted wide attention because of its exceptional classification performance. However, such methods encounter challenges in handling input sample and feature extraction. Regarding the input sample, current hyperspectral image (HSI) classification methods based on spectral–spatial features treat each pixel of the sample equally, resulting in inadequate attention to valuable pixels within 3-D samples. Regarding feature extraction, the classification methods struggle to effectively extract both local and global information from HSIs. Aiming at solving the above problems, we propose the local–global cross fusion network with Gaussian-initialized positional prompting (LGGNet). LGGNet is designed with an end-to-end architecture, primarily comprising the Gaussian-initialized learnable positional prompting and the local–global cross fusion network. The Gaussian-initialized learnable positional prompting introduces prompting technique into HSI classification, utilizing trainable parameters with prior information to learn the spatial importance of different pixels within a sample for the first time. The local–global cross fusion network combines operations such as 3-D convolutional neural network (CNN) feature extraction, transformer feature extraction, and feature fusion, efficiently integrating local and global features. Extensive experiments showcase that LGGNet achieves state-of-the-art performance with limited training samples on four benchmark datasets, all within a lightweight framework. The relevant code is available at https://github.com/ibelieveican2018/LGGNet.**

***Index Terms*— 3-D convolutional neural network (CNN), hyperspectral image (HSI) classification, prompting, remote sensing, transformer.**

I. Introduction

With the rapid development of imaging technology, the spatial and spectral resolution of remote sensing images is constantly improved. Remote sensing image has experienced the development from panchromatic image, multispectral image, and hyperspectral image (HSI). Notably, the HSI contains hundreds of spectral bands with the spectral resolution up to $\lambda/100$. With rich spatial and spectral information of ground objects, HSI plays an important role in natural disaster assessment [1], mineral resources exploration [2], environmental change analysis [3], agricultural monitoring [4], military security [5], and so on.

Predicting a pixel label according to the features contained in each pixel is the task of HSI classification, which has a profound influence on subsequent image processing [6]. As the volume of data to be processed continues to grow, there is a constant refinement in feature acquisition techniques. This evolution has seen HSI classification methods transition from traditional methods to deep learning methods. Categorized by the nature of their features, HSI classification methods based on deep learning can be roughly divided into those based on spectral features [7], spatial feature [8], and spectral–spatial features [9]. Among them, the classification method based on spectral–spatial features has garnered substantial attention due to its ability to effectively utilize the rich the spectral–spatial information of HSIs.

In common HSI classification methods based on spectral–spatial features, the input of the network is generally based on 3-D cube samples. During data division, dense 3-D cube samples are divided around the central pixel. Also, the label of the center pixel is usually treated as the label of the entire 3-D cube sample. However, in the classification prediction process, the predicted value is determined by the characteristics of all pixels within the 3-D cube. In this context, a critical question arises: are all pixels equally vital for predicting the label of the 3-D sample? Surprisingly, there has been limited research into this aspect. Intuitively, pixels close to the center are more likely to share the same class as the center pixel, while those farther away are more likely to differ. This insight implies that not all pixels within a 3-D cube contribute equally; the center pixel holds the most significance in predicting the 3-D sample's label. Ignoring this nuance can lead to surrounding pixels interfering with the central pixel's prediction in a 3-D sample. One potential solution is to devise explicit rules that encode spatial knowledge, such as using Gaussian distribution to reflect prior expertise. However, these artificial rules might not faithfully capture the complexities of HSI classification scenarios. Is there a more adaptive approach

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to learn fine-grained spatial prior knowledge? In this study, we propose prompting as an effective way to automatically learn prior knowledge by merging it with prior knowledge. While standard prompting relies on large-scale pretrained models, such models are lacking in the HSI classification domain. To overcome this deficiency, we employ the Gaussian function with expert-designed prior knowledge to initialize prompting, rather than starting with empty parameters. This strategy compensates for the dearth of prior knowledge, accelerates model convergence, and prevents the model from converging to suboptimal solutions. In addition, we encourage further optimization based on the Gaussian prior, enabling the adaptive learning of the fine-grained importance of spatial pixels within a 3-D sample.

In the classification method based on the spectral–spatial features, exploring effective methods for feature extraction from HSIs is a crucial area for further investigation. Among the array of methods available, the classification method based on 3-D convolutional neural network (CNN) has demonstrated impressive classification performance due to its alignment with the inherent structural characteristics of HSIs. In particular, 3-D CNN-based HSI classification can be categorized into various types, including those based on residual connections [10], [11], attention mechanisms [12], and dense connections [13], [14]. However, these methods solely utilize local information, neglecting the consideration of global features. This limitation results in the inadequate extraction of high-level features present in HSIs. Furthermore, the classification method based on 3-D CNN has a large number of parameters and a long training time, resulting in a waste of resources. More recently, the classification method based on transformer has garnered significant attention. Different from 3-D CNN, transformer excels in capturing the long-term dependence but may struggle to comprehensively extract local features such as lines, corners, and edges within images. Subsequently, a series of transformer-based classification networks, including HSI-BERT [15], SpectralFormer [16], DSS-TRM [17], SSFTT [18], HiT [19], MCT [20], and DHViT [21], have been proposed for HSI classification. Effectively integrating the strengths of the 3-D CNN and transformer to comprehensively capture both local and global information presents an unresolved challenge. In pursuit of this, this article presents a double-branch network comprising 3-D CNN and transformer components. This design aims to leverage both local and global information and introduces a fusion strategy to seamlessly harmonize diverse features, thereby enhancing classification efficacy.

This article introduces LGGNet, a comprehensive framework comprising the Gaussian-initialized learnable positional prompting and the local–global cross fusion network. Specifically, the Gaussian-initialized learnable positional prompting encompasses three crucial stages: Gaussian positional encoding, latent information embedding, and backpropagation parameter updating. This method enhances the attention given to essential pixels within 3-D input samples in the HSI classification based on spectral–spatial features. It achieves this by incorporating trainable vectors infused with prior information. Also, the local–global cross fusion network, a fusion of operations including 3-D CNN feature extraction, transformer feature extraction, and feature fusion, effectively merges local and global features. This involves an initial 3-D feature extraction module, a 3-D spectral–spatial residual module for 3-D CNN feature extraction, a transformer module for transformer feature extraction, and a feature interaction module along with a token cross fusion module for feature fusion. Through the coordination of different components, LGGNet achieves remarkable results, with overall accuracy (OA), average accuracy (AA), and Kappa of LGGNet surpassing 98% even with a minimal 10% training sample ratio on benchmark datasets such a Indian Pines (IP), Kennedy Space Center (KSC), Houston2013 (HU), and Pavia University (PU). Compared with other leading deep learning methods, LGGNet stands out in terms of classification accuracy, computational complexity, parameter numbers, and test time (TT). The main contributions of the article are listed as follows.

1) Gaussian-initialized learnable positional prompting is proposed. This is the first time that prompting is used to solve the problem of insufficient attention given to important pixels within 3-D samples in HSI classification methods based on spectral–spatial features.
2) Local–global cross fusion network is proposed. Comprising a sequence of processes, including 3-D CNN feature extraction, transformer feature extraction, and feature fusion, this network effectively captures both local and global spectral–spatial information from HSIs.
3) LGGNet can achieve state-of-the-art effects using limited input data on four hyperspectral benchmark datasets with lightweight computational complexity.

The remainder of this article is arranged as follows. The related work is introduced in Section II. The methodology of LGGNet is elaborated in Section III. The experimental settings, evaluation results, and corresponding analysis are shown in Section IV. The conclusion and the direction of the future study are illustrated in Section V.

II. Related Work

A. Prompting

Currently, prompting has made remarkable achievements in the field of natural language processing. By incorporating language instructions into the input, prompting enables the pretrained language model to comprehend the task-specific context [22]. This has led to the emergence of prompt tuning, wherein continuous vectors serve as representations and are fine-tuned using gradients for specific tasks [23], [24], [25].

Visual prompting draws inspiration from natural language prompting. For instance, in the training of visual prompting for downstream recognition tasks, a small number of trainable visual prompts based on specific tasks are introduced into the input, and the model backbone is kept frozen [26]. In addition, Bahng et al. [27] adapted this approach for large-scale models by formulating visual prompts at the pixel level.

However, the above prompting methods are predominantly built on pretraining tasks. In the field of HSI processing, there is a lack of large-scale pretrained models. How to use prompting to improve the classification performance of HSIs is a problem worth studying. Under such circumstances,

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the Gaussian-initialized learnable positional prompting is proposed to improve the attention of valuable pixels within the 3-D input sample by incorporating prior positional information in the HSI classification task.

B. Deep Learning Methods for HSI Classification

Deep learning has promoted the development of the classification of HSIs. According to different features, HSI classification methods based on deep learning can be roughly divided into three types: 1) classification methods based on spectral features: these methods use deep belief networks (DBNs) [28], stacked autoencoders (SAEs) [29], recurrent neural networks (RNNs) [7], transformer [15], and so on to obtain spectral features of HSIs; 2) classification methods based on spatial features: such methods generally use CNN [8] to extract spatial features; and 3) classification methods based on spectral–spatial features: these methods can be subdivided into two kinds: one kind is to use 3-D CNN to directly extract spectral–spatial features [30] and another kind is to fuse the spectral features and spatial features together to obtain spectral–spatial features [9]. In the study, the Gaussian-initialized learnable positional prompting is introduced to enhance the attention toward useful pixels within the 3-D cubes for the classification methods based on spectral–spatial features. In addition, the local–global cross fusion network is proposed to combine 3-D CNN and transformer to effectively extract spectral–spatial features. Subsequently, a brief overview of the hyperspectral classification method based on 3-D CNN and transformer is presented.

1) Classification Methods Based on 3-D CNN: The classification method adopting 3-D CNN has garnered substantial attention due to its alignment with the 3-D spectral–spatial cube structure of HSIs [31]. For example, Roy et al. [32] put forward HybridSN, which integrates 2-D CNN and 3-D CNN for HSI classification. Zhang et al. [13] proposed MSDN for classification, which is a multiscale dense model containing horizontal and vertical dimensions. Roy et al. [12] came up with S3EResBOF, which is a network that integrates the residual block, the squeeze-and-excitation module, and the bag-of-feature learning to effectively obtain the channel-wise spectral–spatial features of HSIs. Zhang et al. [10] proposed SSFRAN, which is a network with spectral–spatial fractal residual structure and can capture spectral–spatial features of HSIs effectively. Zhong et al. [11] put forward SSRN, which is a spectral–spatial residual model that can learn the spectral–spatial information of HSIs alternately. Zhang et al. [14] proposed SSDANet, which can effectively obtain the information of HSIs through the spectral–spatial dense connectivity and the spectral–spatial attention mechanism. Paoletti et al. [33] came up with a deep and dense CNN with shortcut connections, which has comparative advantages over other methods.

While the classification method using 3-D CNN significantly enhances classification accuracy and holds an indispensable role in the HSI classification field, it does come with limitations. For instance, it primarily focuses on local information, involves a considerable number of parameters, and requires an extended training time.

2) Classification Methods Based on Transformer: Transformer has gained significant attention in natural language processing due to the ability to obtain long-term dependencies in textual corpus [34], [35]. Recently, vision transformer (ViT) is proposed to deal with visual tasks [36]. Its core concept involves dividing the image into patches and flattening them into 1-D vectors for model input. Notably, transformer has also shown impressive results in various areas, including object tracking [37], image segmentation [38], object detection [39], and image generation [40]. The application of transformer has extended to HSI classification as well.

In the realm of HSI classification, the use of transformer has gained traction. He et al. [15] proposed HSI-BERT, offering flexibility in selecting input regions and employing bidirectional encoder representation from transformer to obtain global dependencies between pixels. Hong et al. [16] proposed a network based on transformer named SpectralFormer to classify hyperspectral data in a sequential way. Liu et al. [17] put forward a deep network named DSS-TRM, which uses a transformer to obtain the features along the spatial and spectral dimensions of HSIs. Sun et al. [18] proposed the SSFTT model, a hybrid network integrating 3-D CNN, 2-D CNN, and transformer to extract spectral–spatial information of HSIs. Yang et al. [19] came up with the HiT, which initially employs a 3-D CNN-based SACP module to capture the spectral–spatial features and subsequently applies the Conv-Permutator module similar to transformer encoder to encode these features. Wang et al. [20] put forward the MCT model, leveraging a multiscale convolution embedding module and transformer module for effective feature extraction through self-supervised pretraining. Xue et al. [21] proposed DHViT, a deep transformer structure composed of spectral, spatial, and LiDAR branches to jointly classify hyperspectral and LiDAR data.

To solve the challenge of effectively acquiring local information with transformer, researchers have undertaken efforts to combine transformer and CNN. However, how to make full use of local–global information and focus on useful information is still a problem worth studying. In view of the above situation, LGGNet is proposed.

III. Methodology

The overall implementation flow of the proposed method is shown in Fig. 1, with the variable annotations provided in Table I. It can be noticed that the input is composed of hyperspectral data $D \in \mathbb{R}^{H\times W\times B}$ and the ground truth y. Preprocessing is initially applied to the hyperspectral data. This involves data standardization operation std$(\cdot)$ on D, transforming the processed data $\tilde{D}$ into a normal distribution. Specifically, the mean is subtracted from D and the result is divided by variance σ to improve the performance of data, as illustrated in the following equation:

$$\tilde{D} = \text{std}(D) = \frac{D-\mu}{\sigma}. \tag{1}$$

Subsequently, PCA is used to reduce the spectral dimension of the $\tilde{D}$ from B to M ($M < B$). This PCA-driven dimensionality

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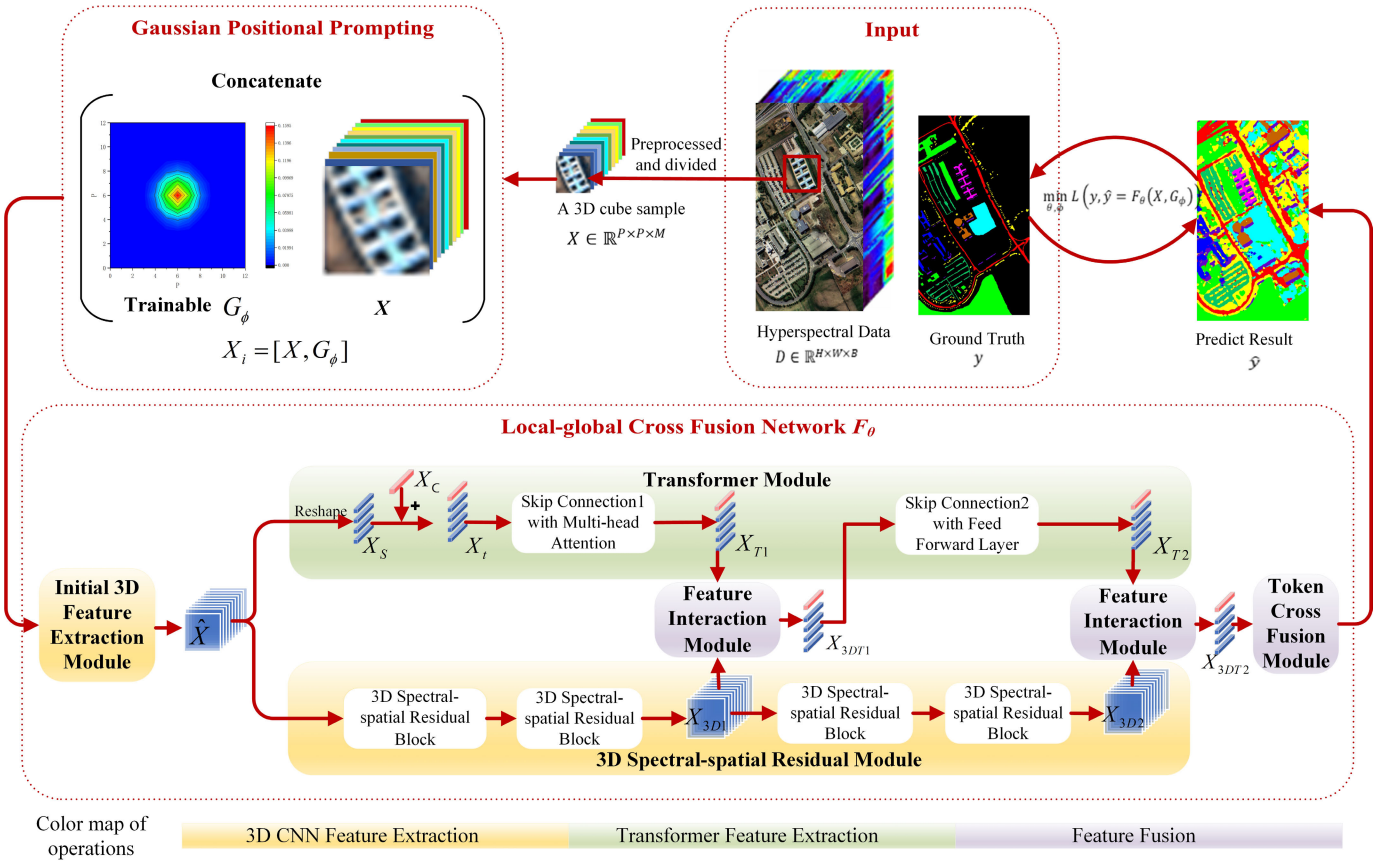


Fig. 1. Overall diagram of LGGNet.

TABLE I
MEANING REPRESENTATION OF THE VARIABLES

Variables	Meaning	Variables	Meaning
$D \in \mathbb{R}^{H\times W\times B}$	Hyperspectral data. Where, H, W, and B represent the dimension of the height, width, and spectrum of HSI.	$X_{Tn}\mathbb{R}^{(\alpha+1)\times\beta}, n = 1,2$	The features obtained from the middle and posterior parts of the Transformer module.
$X \in \mathbb{R}^{P\times P\times M}$	3D cube samples. Where, P indicates the spatial dimension and M represents the spectral dimension of the sample.	$X_{3Dn} \in \mathbb{R}^{\alpha\times\beta}, n = 1,2$	The features obtained from the middle and posterior parts of the 3D spectral-spatial residual module.
$G_\phi \in \mathbb{R}^{P\times P\times 1}$	Learnable prompts. Where, ϕ indicates the parameters to be updated in the prompts.	$X_{3DTn} \in \mathbb{R}^{(\alpha+1)\times\beta}, n = 1,2$	The output of the feature interaction module.
$X_i \in \mathbb{R}^{P\times P\times(M+1)}$	The output X_i of the Gaussian-initialized learnable positional prompting.	y	Ground truth.
$\hat{X}$	The output of the initial 3D feature extraction module.	$\hat{y}$	Predict value.
$X_s \in \mathbb{R}^{\alpha\times\beta}$	The feature token obtained by reshaping. Where, α and β represent the size of the feature token.	F_θ	The local-global cross fusion network. Where, θ indicates the parameters to be updated in the network.
$X_c \in \mathbb{R}^{1\times\beta}$	The class token.	L	The cross-entropy loss function.
$X_t \in \mathbb{R}^{(\alpha+1)\times\beta}$	Features obtained by concatenating the feature token and class token together.		

reduction ensures uniform band processing for varying sensors, resulting in the representation of dimensionality-reduced hyperspectral data as $\tilde{D}_{\text{PCA}} \in \mathbb{R}^{H\times W\times M}$. Since neighborhood pixels are probability the same category, the classification method based on spectral–spatial features usually divides an HSI into 3-D cubes as samples. Each cube's label is determined by its center pixel. Building on this principle, padding is applied to $\tilde{D}_{\text{PCA}}$ to incorporate edge pixel information of the HSI. The data are then divided into N 3-D cube samples, with each represented as $X \in \mathbb{R}^{P\times P\times M}$, where $N = H \times W$, representing sample number of HSI, and P indicates the patch size. Then, the sample set composed of 3-D cubes is split into the training set and the test set in a certain proportion. To solve the problem of the unbalanced training sample number, data balance technology by means of oversampling is also adopted.

Following this, Gaussian-initialized learnable positional prompting is applied to the input sample X. This method concatenates X with trainable prompts G_ϕ that contains prior information. With the iteration of the network, the parameters of G_ϕ are constantly updated. Thus, the network can comprehensively grasp the significance of spatial pixel information within the 3-D sample.

Subsequently, the local–global cross fusion network F_θ is proposed to extract local and global features of HSIs. This network encompasses a sequence of operations, including 3-D CNN feature extraction, transformer feature extraction, and

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feature fusion. Specifically, 3-D CNN feature extraction is carried out through an initial 3-D feature extraction module and a 3-D spectral–spatial residual module. Transformer feature extraction is performed using the transformer module. Also, feature fusion is implemented by the feature interaction module and the token cross fusion module.

In the training process, the gradients of LGGNet are updated through the Adam optimizer, aiming to minimize the cross-entropy loss function L between the ground truth value y and the predicted value $\hat{y}$. For the local–global cross fusion network F_θ with prompts G_ϕ (where X represents the sample), the optimization objective of LGGNet is expressed as follows:

$$\min_{\theta,\phi} L\left(y, \hat{y} = F_\theta\left(X, G_\phi\right)\right) \tag{2}$$

where θ and ϕ indicate the parameters to be updated in the network and prompts, respectively.

A. Gaussian-Initialized Learnable Positional Prompting

In conventional HSI classification models based on deep learning, the label of each input 3-D cube is typically determined by its center pixel. Also, the closer it is to the center, the more likely it is to be in the same category as the center pixel. Inspired by visual prompts that can introduce prior information to the model, we propose the Gaussian-initialized learnable positional prompting. This approach introduces learnable prompts $G_\phi \in \mathbb{R}^{P\times P\times 1}$, which can improve the model's attention to useful pixels of the 3-D input cube by injecting relative or absolute position information into the input $X \in \mathbb{R}^{P\times P\times M}$. The algorithm of Gaussian-initialized learnable positional prompting is shown in Algorithm 1. It mainly consists of three important steps: Gaussian positional encoding to initialize the prompts, latent information embedding to concatenate with the input, and backpropagation parameter updating to optimize the prompts.

Among these steps, the Gaussian positional encoding aims to calculate the importance of each spatial pixel position according to the data characteristics of hyperspectral samples. This yields the initial value $G_{\phi 0}$ of G_ϕ, as shown in the following equations:

$$g(x, y) = \frac{1}{2\pi\sigma^2} e^{-\frac{(x-r)^2+(y-r)^2}{2\sigma^2}} \tag{3}$$

$$\lambda = \sum_{x=0}^{P-1}\sum_{y=0}^{P-1} g(x, y) \tag{4}$$

$$G_{\phi 0} = \frac{g(x, y)}{\lambda} \tag{5}$$

where $r = (P - 1/2)$. σ is the standard deviation, which determines how fast the function decays. λ represents the coefficient of Gaussian positional encoding. The initial value of Gaussian-initialized learnable positional prompts is shown in Fig. 2. It can be noticed that the initial value $G_{\phi 0}$ exhibits large values near the center and small values far from it, aligning with the information distribution characteristics of hyperspectral samples.

The latent information embedding involves concatenating $G_{\phi 0}$ to the input 3-D cube X, serving as input of subsequent layers. This operation introduces the priori spatial information regarding the pixel importance of the input into the model.

Also, in the stage of backpropagation parameter updating, G_ϕ is updated according to the characteristics of the input as the iteration step T increases. These operations enable the proposed model to learn the fine-grained importance of spatial pixel positions. The output X_i of the Gaussian-initialized learnable positional prompting is represented by the following equation:

$$X_i = \left[X, G_{\phi i}\right] = F_\theta\left[X, G_{\phi i-1}\right], \quad i = 1, 2, \ldots, T \tag{6}$$

where $[\cdot, \cdot]$ indicates the concatenation operation and F_θ represents the local–global cross fusion network.

Algorithm 1 Gaussian-Initialized Learnable Positional Prompting

Input: 3-D $sampleX \in R^{P\times P\times M}$, where P represents the spatial dimension and M indicates the spectral dimension of the 3-D sample; the local-global cross fusion network F_θ; the iteration step T.
Output: The output X_i of Gaussian-initialized learnable positional prompting.
1: Radius calculation $r = \frac{P-1}{2}$.
2: **begin**
3: **for** x from 0 to $P - 1$ **do:**
4: **for** y from 0 to $P - 1$ **do:**
5: The values of each spatial position are calculated according to (3)–(5), and the initial value $G_{\phi 0} \in \mathbb{R}^{P\times P}$ of prompts G_ϕ is obtained.
6: **end**
7: Perform the operation of latent information embedding on the input X and $G_{\phi 0}$.
8: **begin**
9: **for** the iteration from 1 to T **do:**
10: Backpropagation parameter updating is implemented, and the output X_i is shown as (6).
11: **end**

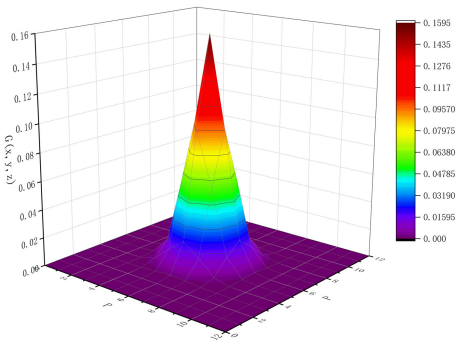


Fig. 2. Initial value of Gaussian-initialized learnable positional prompts.

B. Three-Dimensional CNN Feature Extraction

1) Initial 3-D Feature Extraction Module: The traditional ViT divides the image into 2-D patches, typically 16×16 or 32×32 in size, and the patches are subsequently flattened into 1-D vectors for transformer input [36]. However, this

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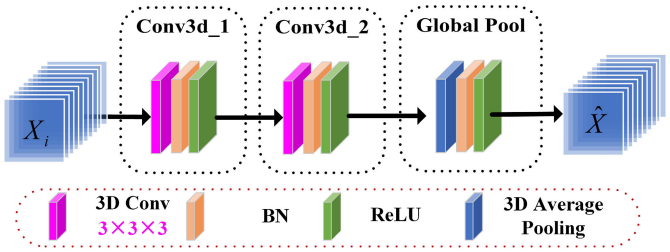


Fig. 3. Schematic of initial 3-D feature extraction module.

means of data processing is ill-suited for HSIs. Different from natural images, hyperspectral remote sensing images contain rich spatial and spectral information. Flattening these images disrupts their inherent structure, thus hindering the complete capture of local information. To solve the above problems, the initial 3-D feature extraction module is adopted to get the low-level local spectral–spatial semantic information of HSIs.

The structure of the initial 3-D feature extraction module is straightforward, as shown in Fig. 3. The input X_i obtained by Gaussian-initialized learnable positional prompting undergoes through two Conv3d layers and a global pooling layer to obtain low-level features $\hat{X}$. Specifically, Conv3d layers employ 3-D convolution with a kernel size of $3 \times 3 \times 3$, accompanied by batch normalization (BN) and ReLU. The global pooling layer involves 3-D average pooling, BN, and ReLU.

2) Three-Dimensional Spectral–Spatial Residual Module: The 3-D spectral–spatial residual module, comprising four stacked 3-D spectral–spatial residual blocks, is designed to capture the high-level local spectral–spatial information from HSIs. As shown in Fig. 1, the initial 3-D feature extraction module's output, denoted as $\hat{X}$, serves as the input to the 3-D spectral–spatial residual module. Following the first two 3-D spectral–spatial residual blocks, the output X_{3D1} is obtained, and subsequent to the last two 3-D spectral–spatial residual blocks, the output X_{3D2} is produced.

Each 3-D spectral–spatial residual block comprises a 3-D spatial residual block and a 3-D spectral residual block, as shown in Fig. 4. Suppose that input X comprises r feature maps with the size of $u \times u \times v$, and it undergoes the 3-D spatial residual block to yield X_{spatial}. Subsequently, X_{spatial} passes through the 3-D spectral residual block to generate X_{ss}, which retains the same dimensions as input X. In detail, within 3-D spatial residual block, input X undergoes the composite function $F_{\text{spatial}}(\cdot)$, which is constituted by a series of operations, to derive $F_{\text{spatial}}(X)$. Then, $F_{\text{spatial}}(X)$ is added to input X and passed through the ReLU $\varphi(\cdot)$ to improve the nonlinear expression ability. By combining the features of different levels through skip connections, the model performance can be heightened. The 3-D spatial residual block is represented by the following equation:

$$X_{\text{spatial}} = \varphi\big(X + F_{\text{spatial}}(X)\big). \tag{7}$$

The difference between 3-D spatial residual block and spectral residual block lies in their respective composite functions. The composite functions in spectral residual block are denoted as $F_{\text{spectral}}(\cdot)$, and the correlation calculation of 3-D spectral–spatial residual block is shown in the following equation:

$$X_{\text{ss}} = \varphi\big(X_{\text{spatial}} + F_{\text{spectral}}\big(X_{\text{spatial}}\big)\big). \tag{8}$$

Both $F_{\text{spatial}}(\cdot)$ and $F_{\text{spectral}}(\cdot)$ comprise 3-D convolution, BN, Dropout, and ReLU. The discrepancy between the two lies in the kernel size employed in 3-D convolution. The 3-D spatial residual block adopts the kernel size $3 \times 3 \times 1$, while the 3-D spectral residual block adopts the kernel size $1 \times 1 \times 3$. By connecting the 3-D spatial residual block and 3-D spectral residual block, the 3-D spectral–spatial residual module continuously extracts both spectral information and spatial content from HSIs.

C. Transformer Feature Extraction

To capture the intrinsic relationship and long-term dependence within the data, the transformer module with the structure similar to the transformer encoder is adopted. In the transformer module, the input $\hat{X}$ is reshaped into 1-D vector named feature token $X_s \in \mathbb{R}^{\alpha \times \beta}$. Then, the class tokens $X_c \in \mathbb{R}^{1 \times \beta}$ is appended to the feature token as the input $X_t \in \mathbb{R}^{(\alpha+1) \times \beta}$ for the subsequent operations. The transformer module mainly consists of layer normalization, multihead attention, and feedforward layer. In addition, the skip connection is used to counteract the vanishing gradient problem. The feature obtained by the skip connection1 with multihead attention is denoted as X_{T1}, while the features obtained by the skip connection2 with feedforward layer are denoted X_{T2}. Both the multihead attention and the feedforward layer are essential parts of transformer.

The multihead attention is constructed by the integration of self-attention layers, as shown in Fig. 5. Initially, the input token is mapped into the query–key–value (QKV) space. Subsequently, the attention score ξ between the vector Q and the vector K is computed using inner product operations. Following this, ξ is divided by the scaling factor $(\delta)^{1/2}$ to distract the attention of the value with high attention scores. This results in the normalized attention score $\tilde{\xi}$ that is obtained. Next, the Softmax function is applied to map $\tilde{\xi}$ into the range (0, 1). Ultimately, vector V is multiplied by $\tilde{\xi}$, increase attention toward the current pixel while reducing attention to irrelevant content. Through the above operation, the output Z of the self-attention is generated, as illustrated in the following equation:

$$\begin{aligned} Z &= \text{Attention}(Q, K, V) \\ &= \text{softmax}\big(\tilde{\xi}\big) \times V = \text{softmax}\left(\frac{QK^T}{\sqrt{\delta}}\right) \times V. \end{aligned} \tag{9}$$

The feedforward layer is quite simple, which is composed of two fully connected layers and the activation function GELU. This structure enhances the representation capacity of the model.

D. Feature Fusion

1) Feature Interaction Module: To fully fuse the local information and global information of HSIs, the feature interaction module is proposed. This module is adopted in the middle and postprocessing part of the local–global cross fusion network.

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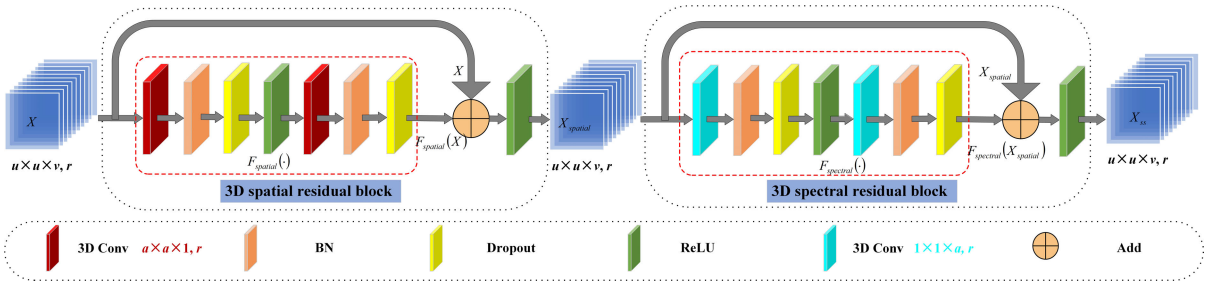


Fig. 4. Structure of 3-D spectral–spatial residual block.

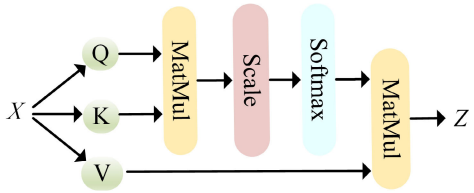


Fig. 5. Structure of self-attention.

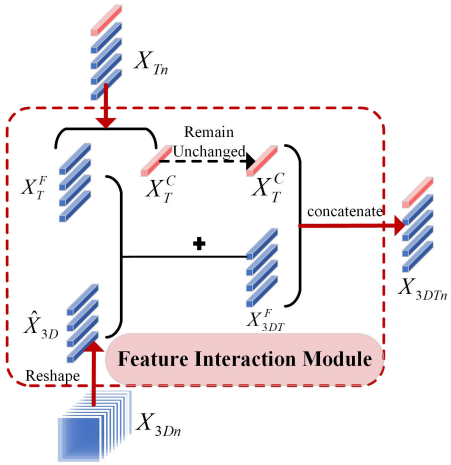


Fig. 6. Structure of the feature interaction module.

The structure of feature interaction module is shown in Fig. 6. Initially, the feature $X_{Tn} \in \mathbb{R}^{(\alpha+1)\times\beta}, n = 1, 2$, of the transformer branch is split into the class token $X_T^C \in \mathbb{R}^{1\times\beta}$ and the feature token $X_T^F \in \mathbb{R}^{\alpha\times\beta}$. Here, X_T^C and X_T^F contain the category and feature information extracted by transformer, respectively. Notably, although transformer exhibits exceptional aptitude for global feature extraction, its capacity for local feature extraction remains constrained. As a result, X_T^F may not comprehensively capture the low-level information within HSIs. To address this challenge, the feature $X_{3Dn} \in \mathbb{R}^{\alpha\times\beta}$, corresponding to $n = 1, 2$, obtained from the 3-D spectral–spatial residual module, is reshaped and added to X_T^F, thus yielding the fully enriched local spectral–spatial feature $X_{3DT}^F \in \mathbb{R}^{\alpha\times\beta}$. Subsequently, X_{3DT}^F is concatenated with X_T^c to generate the ultimate output $X_{3DTn} \in \mathbb{R}^{(\alpha+1)\times\beta}$, where $n = 1, 2$.

2) Token Cross Fusion Module: The conventional ViT exclusively relies on class token for classification, leading to the potential loss of valuable feature information. To address this limitation, the token cross fusion module, which can integrate class token and feature token effectively, is proposed, as shown in Fig. 7.

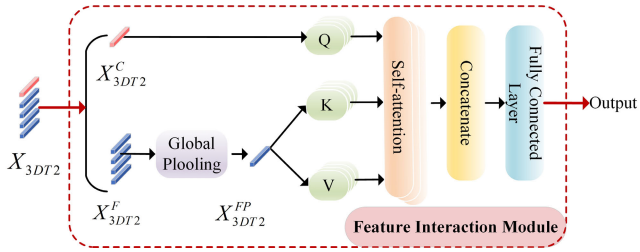


Fig. 7. Structure of the token cross fusion module.

Within the token cross fusion module, the input $X_{3DT2} \in \mathbb{R}^{(\alpha+1)\times\beta}$ is partitioned into the class token $X_{3DT2}^C \in \mathbb{R}^{1\times\beta}$ and feature token $X_{3DT2}^F \in \mathbb{R}^{\alpha\times\beta}$. Subsequently, the global pooling layer is applied to X_{3DT2}^F, yielding the feature $X_{3DT2}^{FP} \in \mathbb{R}^{1\times\beta}$. Importantly, X_{3DT2}^C and X_{3DT2}^{FP} have the same dimension. Following this, X_{3DT2}^C and X_{3DT2}^{FP} are input into the attention mechanism for feature fusion, where X_{3DT2}^C is mapped as query Q and X_{3DT2}^{FP} is mapped as key K and value V. Ultimately, the fully connected layer is employed to compute the probability distribution across various categories.

E. Implementation Details

To gain further insight into the implementation details of LGGNet, we present a concise representation of the input and output feature map sizes for each module, as shown in Table II. Specifically, the Gaussian-initialized learnable positional prompting takes input as a 3-D HSI sample with the dimension of $1 \times 15 \times 13 \times 13$. It concatenates this input with learnable prompts to generate an output of size $1 \times 16 \times 13 \times 13$. Subsequently, this output undergoes the initial 3-D feature extraction module, comprising two Conv3d layers and a global pooling layer, yielding a feature map of the dimension $12 \times 8 \times 4 \times 4$. These obtained feature maps are individually fed into the 3-D spectral–spatial residual module and the transformer module, generating feature maps of sizes $12 \times 8 \times 4 \times 4$ and 17×96, respectively. To facilitate interaction and alignment of features within the 3-D spectral–spatial residual module and transformer module, we introduce the feature interaction module, ultimately yielding the feature map of size 17×96. Finally, the token cross fusion module is employed to effectively merge class token and feature token, ultimately obtaining features with the size equal to the class number of the HSI.

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TABLE II
INPUT AND OUTPUT FEATURE MAP SIZES FOR DIFFERENT MODULES IN LGGNET

Module	Input Size	Output Size
Gaussian-initialized Learnable Positional Prompting	$1 \times 15 \times 13 \times 13$	$1 \times 16 \times 13 \times 13$
Initial 3D Feature Extraction Module	$1 \times 16 \times 13 \times 13$	$12 \times 8 \times 4 \times 4$
3D Spectral-spatial Residual Module	$12 \times 8 \times 4 \times 4$	$12 \times 8 \times 4 \times 4$
Transformer Module	$12 \times 8 \times 4 \times 4$	17×96
Feature Interaction Module	$12 \times 8 \times 4 \times 4$ (from 3D Spectral-spatial Residual Module); 17×96 (from Transformer Module)	17×96
Token Cross Fusion Module	17×96	class number

IV. EXPERIMENTS, EVALUATION, AND ANALYSIS

In this section, experimental datasets and experimental settings are elaborated first. Subsequently, experimental results are evaluated quantitatively and qualitatively. Following the evaluation, the impact of ablation studies, Gaussian-initialized learnable positional prompting, patch size, band number, and training sample number on the experimental results are analyzed. Details are presented as follows.

A. Experiments

1) Experimental Datasets: Experiments were carried out on four benchmark datasets of IP, KSC, HU, and PU to verify the generalization capability of LGGNet.

The IP dataset was captured by the AVIRIS hyperspectral imager in June 1992, covering northwest Indiana. It includes 16 land-cover categories. In the spatial dimension, it includes 145×145 pixels and has a 20-m spatial resolution. In the spectral dimension, it is made up of 224 spectral bands, in the range between 400 and 2500 nm. The corrected IP dataset remains 200 spectral bands after discarding the bands polluted by water vapor absorption.

The KSC dataset was acquired by AVIRIS in March 1996 and photographed scenes in Florida. It contains 13 land-cover categories. In the spatial dimension, it contains 512×614 pixels with the 18-m spatial resolution. In the spectral dimension, it includes 224 spectral bands, in the range between 400 and 2500 nm. The corrected KSC dataset retains 176 bands after abandoning the noise bands.

The HU dataset[1] was captured by the CASI sensor and released in 2013, covering regions of Houston University and the surrounding area. It contains 15 categories. In the spatial dimension, it includes 349×1905 pixels with the 2.5-m spatial resolution. In the spectral dimension, it contains 144 spectral bands, in the range between 380 and 1050 nm.

The PU dataset was captured by the spectral imager ROSIS in 2021, depicting scenes in North Italy. It contains nine types of urban land-cover categories. In the spatial dimension, it includes 610×340 pixels with the 1.3-m spatial resolution. In the spectral dimension, it contains 115 spectral bands, in the range between 430 and 860 nm. After removing noise spectral bands, 103 bands are retained in the corrected PU dataset.

2) Experimental Settings: The experiments were carried out on the server with the graphic processing unit of NVIDIA Tesla V100S. The programming language used for the experiments was Python. The deep learning framework PyTorch 1.7.1 was employed for implementing LGGNet. In addition, to compare with other methods, Tensorflow1.13.1 and Keras2.1.0 frameworks were also employed in this study.

During the experiment, the Adam optimizer was employed for gradient descent optimization. L2 regularization was adopted to mitigate overfitting, with a decay weight of 0.0001. The learning rate and iteration were set to 0.0001 and 5000, respectively. Moreover, the training dataset size was set to 10% of the total samples. The band number and the patch size were set to 15 and 13, respectively. Across all datasets, a batch size of 512 was used.

OA, AA, and Kappa coefficient (κ) were utilized to evaluate the precision of the model. Besides, the number of parameters (Param), floating-point operations (FLOPs), and TT were adopted to evaluate the size of the network. In summary, the study evaluates the classification model from both accuracy and complexity perspectives.

B. Evaluation

This article compares LGGNet with other deep learning methods to demonstrate the superiority of LGGNet. The comparison methods are: HybridSN [32], MSDN [13], S3EResBOF [12], SSFRAN [10], SSRN [11], HSI-BERT [15], and SSFTT [18]. These methods adopt different means of implementation. In terms of the technology adopted, HybridSN uses 2-D CNN and 3-D CNN for classification; MSDN, S3EResBOF, SSFRAN, and SSRN adopt 3-D CNN for classification; and HSI-BERT and SSFTT are based on transformer for classification. Regarding the frameworks used, HybridSN, MSDN, S3EResBOF, and SSRN are implemented using the Keras framework; SSFRAN and HSI-BERT are implemented using the TensorFlow framework; and SSFTT is implemented using the Pytorch framework.

For a fair comparison, the training sample ratio is set to 10% for each method. Quantitative and qualitative evaluations of different comparison methods are carried out as follows.

1) Quantitative Evaluation: This article compares the results of different classification methods on the datasets of IP, KSC, HU, and PU. To eliminate the influence of randomness, five Monte Carlo runs are conducted for each method, and the results represented by mean and variance are shown in Tables III–VI. The optimal results are highlighted in bold.

The following conclusions can be drawn from the tables.

a) The proposed LGGNet consistently achieves the highest OA, AA, and Kappa compared with other methods on the datasets of IP, KSC, and HU. Even on the PU dataset, where it falls slightly short in terms of accuracy, LGGNet still achieves an impressive accuracy level exceeding 99%. From the perspective of classification precision and generalization, LGGNet establishes itself as a state-of-the-art approach.

[1]2013 IEEE GRSS Data Fusion Contest, Online: http://www.grss-ieee.org/community/technical-committees/data-fusion/

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TABLE III
QUANTITATIVE RESULTS ON THE IP DATASET

Class Name	Samples	HybridSN	MSDN	S3EResBOF	SSFRAN	SSRN	HSI-BERT	SSFTTNet	LGGNet
Asphalt	46	93.66±51.39	99.13±3.03	95.33±14.63	97.14±5.44	**100.00±0.00**	44.88±48.97	95.61±22.39	**100.00±0.00**
Corn-notill	1428	94.49±0.33	81.85±20.19	96.57±6.08	94.40±3.84	**96.97±16.11**	95.94±2.72	93.59±6.98	95.78±0.41
Corn-mintill	830	98.20±0.60	80.57±18.78	97.01±2.46	92.50±7.33	97.53±3.18	95.93±5.06	98.63±0.87	**99.59±0.12**
Corn	237	96.06±6.75	82.48±140.97	97.95±9.35	97.20±1.13	93.65±2.51	88.45±14.87	95.40±5.06	**99.44±0.57**
Grass-pasture	483	97.88±5.26	96.97±1.60	97.78±0.92	96.73±0.85	95.16±1.29	97.05±3.62	98.57±1.30	**99.91±0.03**
Grass-trees	730	98.24± 0.40	90.91±48.72	97.89±0.24	98.08±2.10	**99.60±0.06**	99.48±0.14	97.59±3.57	99.20±0.07
Grass-pasture-mowed	28	**100.00±0.00**	**100.00±0.00**	76.16±53.66	90.77±151.47	0.00±0.00	59.20±213.76	98.40±3.84	**100.00±0.00**
Hay-windrowed	478	**100.00±0.00**	97.18±1.10	99.89±0.04	**100.00±0.00**	97.45±1.56	**100.00±0.00**	99.68±0.10	99.95±0.01
Oats	20	95.55±17.28	**100.00±0.00**	98.33±11.10	58.89±229.65	0.00±0.00	43.33±91.36	70.00±143.20	98.89±4.95
Soybean-notill	972	**98.99±0.10**	92.39±9.31	97.80±2.65	93.97±5.93	98.75±3.34	95.95±1.71	97.19±1.83	98.08±0.44
Soybean-mintill	2455	**99.35±0.14**	85.69±11.13	97.01±1.73	96.78±0.68	97.22±5.36	98.65±0.34	98.94±0.26	99.07±0.07
Soybean-clean	593	95.73±6.56	81.08±21.96	95.63±1.32	92.43±7.94	98.14±1.54	92.16±7.21	94.91±4.14	**98.24±0.51**
Wheat	205	98.38±3.86	96.13±5.63	96.80±7.33	99.57±0.16	99.66±0.47	99.46±0.24	97.39±1.59	**99.78±0.07**
Woods	1265	99.39±0.22	95.93±1.30	97.53±0.47	99.51±0.29	99.48±0.04	99.79±0.02	99.67±0.05	**100.00±0.00**
Building-Grass-Trees-Drives	386	97.18±2.23	87.74±14.23	97.21±4.24	92.47±22.93	97.58±6.94	93.66±4.02	97.91±2.99	**99.71±0.07**
Stone-Steel-Towers	93	85.72±48.78	95.17±15.49	92.26±41.16	94.29±21.22	**96.53±1.35**	95.18±29.61	86.99±131.41	93.97±13.95
κ(%)		97.59±0.27	86.03±0.82	96.69±0.39	95.38±0.17	97.37±3.43	96.25±0.20	97.09±0.43	**98.54±0.01**
OA(%)		97.89±0.21	87.82±0.61	97.10±0.30	95.96±0.13	97.70±2.61	96.71±0.15	97.45±0.33	**98.72±0.01**
AA(%)	Total:	96.80±1.34	91.45±0.68	95.70±0.72	93.42±1.86	85.48±0.70	87.44±1.89	95.03±2.46	**98.85±0.05**
FLOPs(MMac)	10249	76.83±0.00	18.63±0.00	6.37±0.00	280.35±0.00	7.27±0.00	7.49±0.00	11.40±0.00	**5.56±0.00**
Param(M)		15.37±0.00	3.13±0.00	0.74±0.00	40.05±0.00	0.69±0.00	0.39±0.00	0.15±0.00	**0.09±0.00**
TT(s)		1.92± 0.01	19.86±3.66	2.53±0.02	22.97±0.05	10.34±0.39	6.46±0.11	0.60±0.01	**0.23±0.00**

TABLE IV
QUANTITATIVE RESULTS ON THE KSC DATASET

Class Name	Samples	HybridSN	MSDN	S3EResBOF	SSFRAN	SSRN	HSI-BERT	SSFTTNet	LGGNet
Scrub	761	95.71±8.52	99.14±1.24	90.37±40.46	99.09±0.16	99.94±0.02	**100.00±0.00**	98.71±0.45	**100.0±0.00**
Willow swamp	243	72.97±13.78	92.26±29.34	87.57±12.99	86.57±85.42	98.74±1.44	97.16±1.47	49.27±121.19	**99.72±0.31**
CP hammock	256	91.91±7.53	82.09±5.99	76.44±265.99	91.52±45.02	**99.63±0.12**	91.48±19.58	31.09±299.98	98.48±7.95
Slash pine	252	60.53±91.68	91.81±19.45	85.51±110.71	87.05±23.71	**99.82±0.05**	74.96±13.20	35.84±21.22	97.00±6.33
Oak/Broadleaf	161	98.21±2.97	84.88±19.08	96.00±24.12	95.58±5.45	**99.57±0.75**	73.47±47.94	40.69±244.43	97.91±3.68
Hardwood	229	82.72±20.99	80.28±24.48	94.64±22.69	95.46±17.31	99.78±0.19	94.85±19.29	37.09±438.71	**100.0±0.00**
Swamp	105	75.75±58.55	96.60±21.73	99.07±3.46	98.74±2.40	94.31±15.88	98.72±4.26	40.85±699.69	**100.0±0.00**
Graminoid marsh	431	90.52±27.14	82.24±24.96	76.11±403.92	97.58±3.21	98.28±0.24	99.74±0.03	71.74±110.72	**100.0±0.00**
Spartina marsh	520	73.89±29.22	86.44 +- 8.92	71.75±333.02	96.80±0.88	99.87±0.01	**100.00±0.00**	90.06±15.83	99.9±0.01
Cattail marsh	404	75.39±22.15	93.77±4.64	89.05±176.59	98.79±0.65	98.37±1.24	**100.00±0.00**	77.14±24.19	**100.0±0.00**
Salt marsh	419	98.78±1.50	99.77±0.08	99.82±0.13	98.36±0.37	97.86±1.87	**100.00±0.00**	95.29±0.82	**100.0±0.00**
Mud flats	503	93.16±2.67	96.07±4.18	96.19±2.41	98.59±0.64	**100.00±0.00**	99.47±1.12	79.11±49.02	**100.0±0.00**
Water	927	99.88±0.06	97.79±0.37	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	**100.00±0.00**	99.98±0.00	**100.0±0.00**
κ(%)		87.09±2.93	91.49±0.32	85.70±45.74	96.61±0.59	99.21±0.05	96.77±0.08	74.87±10.95	**99.69±0.03**
OA(%)		88.42±2.35	92.36±0.26	87.19±36.69	96.95±0.47	99.29±0.04	97.10±0.06	77.65±8.27	**99.72±0.02**
AA(%)	Total:	85.34±3.62	91.01±0.54	89.43±16.34	95.70±0.98	98.94±0.15	94.60±0.13	65.14±29.54	**99.46±0.05**
FLOPs(MMac)	5211	72.68±0.00	31.04±0.00	5.77±0.00	265.63±0.00	6.49±0.00	6.76±0.00	11.40±0.00	**5.55±0.00**
Param(M)		14.54±0.00	5.2±0.00	0.68±0.00	37.94±0.00	0.62±0.00	0.36±0.00	0.15±0.00	**0.09±0.00**
TT(s)		0.75±0.00	23.72±0.11	0.87±0.00	12.01±0.03	11.91±0.11	2.84±0.00	0.29±0.00	**0.12±0.00**

b) LGGNet stands out with its minimal number of FLOPs across IP, KSC, and PU. While the HU dataset witnesses SSRN with the highest FLOPs, it lags significantly behind LGGNet in terms of classification accuracy. Moreover, LGGNet not only excels in model size but also maintains the lead in terms of the number of parameters and TT compared to comparative methods. Specifically, due to the effective fusion of 3-D CNN and transformer, LGGNet achieves superior accuracy with a mere 0.09M parameters across four experimental datasets. Fewer parameters naturally result in faster testing times. Therefore, when considering model dimensions, LGGNet emerges as a state-of-the-art choice.

c) LGGNet is still able to achieve satisfactory results even with limited labeled samples. With training samples constituting only 10% of the total, the experimental results validate this claim. On the datasets with a scarcity of samples, such as IP, KSC, and HU, LGGNet has reached the state-of-the-art level. It still manages to achieve excellent results even on the PU dataset with a larger number of input samples. In addition, LGGNet has a strong discriminant ability for the category with few samples, such as grass-pasture-mowed in the IP dataset, swamp in the KSC dataset, water in the HU dataset, and bitumen in the PU dataset.

d) LGGNet's success can be attributed to its key components: Gaussian-initialized learnable positional prompting and the local–global cross fusion network.

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TABLE V
QUANTITATIVE RESULTS ON THE HU DATASET

Class Name	Samples	HybridSN	MSDN	S3EResBOF	SSFRAN	SSRN	HSI-BERT	SSFTTNet	LGGNet
Healthy grass	1251	98.49±0.47	95.61±0.67	**99.86±0.01**	97.80±0.58	99.71±0.03	98.59±0.91	98.66±0.02	99.65±0.04
Stressed grass	1254	99.89±0.02	96.89±0.05	**99.98±0.00**	98.35±0.46	99.54±0.20	99.93±0.00	99.70±0.05	99.95±0.00
Synthetic grass	697	99.94±0.01	97.18±11.40	99.58±0.71	99.78±0.03	**100.00±0.00**	99.97±0.00	99.49±0.18	99.86±0.00
Trees	1244	98.82±0.82	99.01±0.62	99.13±0.70	98.71±0.50	99.83±0.03	98.77±0.70	96.76±3.53	**99.85±0.02**
Soil	1242	**100.00±0.00**	98.18±0.59	99.33±0.64	99.12±0.24	99.85±0.02	**100.00±0.00**	99.93±0.02	**100.00±0.00**
Water	325	97.60±0.14	92.11±22.65	99.76±0.23	93.45±2.54	**100.00±0.00**	91.37±11.18	99.32±0.52	**100.00±0.00**
Residential	1268	96.69±0.97	95.14±2.09	98.57±0.53	98.49±0.22	**99.42±1.15**	98.51±0.63	95.64±1.63	97.58±0.35
Commercial	1244	98.41±0.71	93.33±10.51	**99.86±0.08**	92.23±4.72	99.36±0.55	97.11±1.58	96.52±0.44	98.41±0.27
Road	1252	96.11±0.54	91.46±0.57	99.03±0.28	96.15±0.77	**99.16±0.12**	96.98±0.81	98.67±0.45	98.46±0.08
Highway	1277	99.91±0.02	92.11±1.10	99.17± 0.62	96.56±2.39	99.26±0.02	99.89±0.01	99.31±0.08	**100.00±0.00**
Railway	1235	99.06±0.96	96.47±1.73	99.78±0.03	92.43±1.13	98.86±1.31	98.85±0.80	99.62±0.16	**99.94±0.01**
Parking lot 1	1233	**99.64±0.00**	93.45±3.42	99.36±0.61	93.49±1.18	98.66±0.15	98.47±0.47	98.17±5.31	99.52±0.01
Parking lot 2	469	97.20±2.91	94.01±1.64	**99.17±0.65**	96.31±2.97	98.99±0.53	94.12±1.20	96.67±0.83	98.79±0.80
Tennis court	428	**100.00±0.00**	98.20±3.48	97.99±8.35	99.84±0.02	99.78±0.19	99.84±0.02	99.58±0.23	**100.00±0.00**
Running track	660	99.49±0.31	98.16±1.76	98.86±3.68	99.83±0.07	99.89±0.02	**100.00±0.00**	98.96±0.80	99.89±0.01
κ(%)		98.66±0.03	94.93±0.16	99.27±0.01	96.43±0.17	99.31±0.02	98.43±0.08	98.25±0.13	**99.34±0.00**
OA(%)		98.76±0.03	95.31±0.14	99.33±0.01	96.69±0.15	99.37±0.01	98.55±0.07	98.38±0.11	**99.39±0.00**
AA(%)	Total:	98.75±0.02	95.42±0.43	99.28±0.02	96.84±0.09	99.43±0.02	98.21±0.14	98.47±0.10	**99.46±0.00**
FLOPs(MMac)	15029	72.68±0.00	25.69±0.00	5.77±0.00	236.29±0.00	**5.46±0.00**	7.16±0.00	11.40±0.00	5.5±0.00
Param(M)		14.54±0.00	4.29±0.00	0.68±0.00	33.75±0.00	0.52±0.00	0.38±0.00	0.15±0.00	**0.09±0.00**
TT(s)		1.97±0.02	118.39±21.10	2.59±0.02	23.12±0.08	63.84±7.48	6.68±0.03	0.75±0.00	**0.33±0.00**

TABLE VI
QUANTITATIVE RESULTS ON THE PU DATASET

Class Name	Samples	HybridSN	MSDN	S3EResBOF	SSFRAN	SSRN	HSI-BERT	SSFTTNet	LGGNet
Asphalt	6631	99.91±0.01	98.97±0.30	99.65±0.05	99.75±0.01	99.41±0.29	**99.98±0.00**	99.85±0.01	**99.98±0.00**
Meadows	18649	99.99±0.00	99.85±0.01	99.92±0.01	99.35±0.10	98.41±2.10	99.97±0.00	99.93±0.01	**100.00±0.00**
Gravel	2099	99.19±0.14	98.41±0.52	95.66±19.98	98.56±0.68	91.49±62.67	**99.69±0.05**	98.73±0.56	99.54±0.05
Trees	3064	98.93±0.15	**99.86±0.06**	98.50±0.81	99.27±0.25	99.11±1.20	99.61±0.02	98.68±0.11	99.67±0.01
Painted metal sheets	1345	99.93±0.01	99.84±0.01	99.63±0.10	99.98±0.00	99.91±0.03	**100.00±0.00**	99.14±1.51	99.96±0.00
Bare Soil	5029	**100.00±0.00**	99.84±0.01	99.88±0.02	99.84±0.02	98.53±1.00	**100.00±0.00**	99.92±0.02	**100.00±0.00**
Bitumen	1330	99.87±0.05	99.45±0.38	99.88±0.02	99.61±0.11	97.90±6.75	99.98±0.00	99.82±0.01	**100.00±0.00**
Self-Blocking Bricks	3682	99.54±0.02	97.79±0.52	99.25±0.54	99.53±0.06	95.97±7.81	**99.87±0.01**	98.78±1.34	99.55±0.02
Shadows	947	94.74±1.25	99.77±0.06	96.87±17.43	99.86±0.05	99.80±0.11	**99.88±0.01**	97.58±0.81	97.85±0.15
κ(%)		99.60±0.00	99.28±0.01	99.21±0.13	99.31±0.06	97.43±1.88	**99.90±0.00**	99.46±0.07	99.82±0.00
OA(%)		99.70±0.00	99.45±0.01	99.40±0.07	99.47±0.03	98.07±1.05	**99.93±0.00**	99.59±0.04	99.86±0.00
AA(%)	Total:	99.12±0.02	99.31±0.02	98.80±0.51	99.53±0.02	97.83±1.68	**99.89±0.00**	99.16±0.07	99.62±0.00
FLOPs(MMac)	42776	72.67±0.00	25.63±0.00	5.77±0.00	192.17±0.00	**4.17±0.00**	5.71±0.00	11.40±0.00	5.55±0.00
Param(M)		14.53±0.00	4.29±0.00	0.68±0.00	27.45±0.00	0.39±0.00	0.30±0.00	0.15±0.00	**0.09±0.00**
TT(s)		5.15±0.43	93.52±22.20	7.25±0.09	121.22±87.70	28.67±0.12	14.48±0.17	2.63±0.16	**1.03±0.00**

Gaussian-initialized learnable positional prompting learns the spatial importance of different pixels in a sample by introducing trainable parameters with prior information. Also, the local–global cross fusion network fuses 3-D CNN and transformer together, enabling comprehensive extraction of local and global information, resulting in remarkable classification performance while maintaining a lightweight architecture.

2) Qualitative Evaluation: This article compares the visualization maps of different classification methods on four benchmark datasets, as shown in Figs. 8–11. As different deep learning comparison methods have achieved high classification performance, there are slight differences in the overall classification maps. For a more detailed analysis, local aspects of the classification maps are magnified.

From the observations of these visualization maps, the following conclusions can be drawn.

a) Among different comparison methods, the classification map of LGGNet has the least noise, the smoothest boundaries, the least misclassifications, and the highest level of agreement with the ground truth map. These observations highlight the exceptional classification performance of LGGNet.

b) Compared with other classification methods, LGGNet demonstrates a satisfactory visual classification effect on four experimental datasets. While certain comparison methods excel in classification on individual datasets, they lack the generalization capability to perform consistently across all datasets. Through these analyses, the superior generalization ability of LGGNet is evidenced.

C. Analysis

To further explore the classification performance of LGGNet on experimental datasets, an exploration of various factors is undertaken. These factors include ablation studies, the impact of Gaussian-initialized learned positional prompting, patch size, band number, and training sample number on the performance of LGGNet.

1) Analysis of the Ablation Study: LGGNet is primarily composed of two key components: Gaussian-initialized

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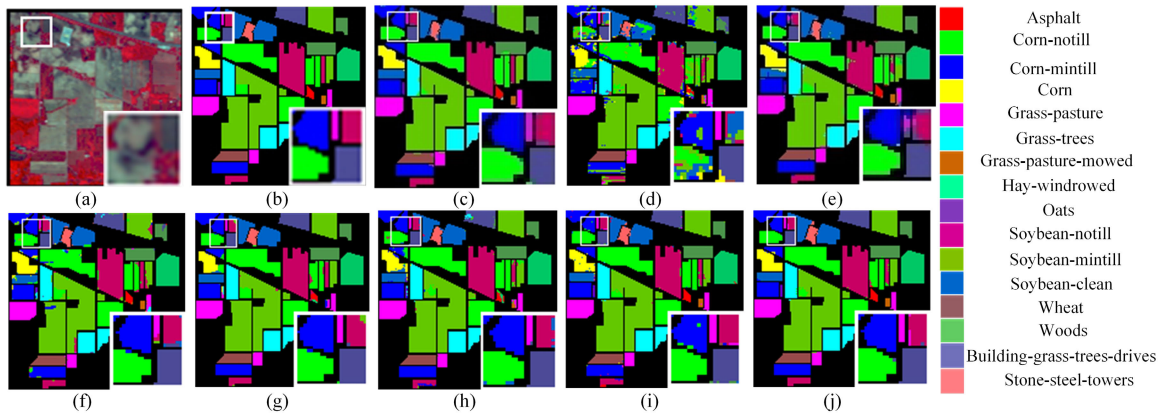


Fig. 8. Visualization maps of comparative methods on the IP dataset: (a) false-color map, (b) ground truth map, (c) HybridSN, (d) MSDN, (e) S3EResBOF, (f) SSFRAN, (g) SSRN, (h) HSI-BERT, (i) SSFTT, and (j) LGGNet.

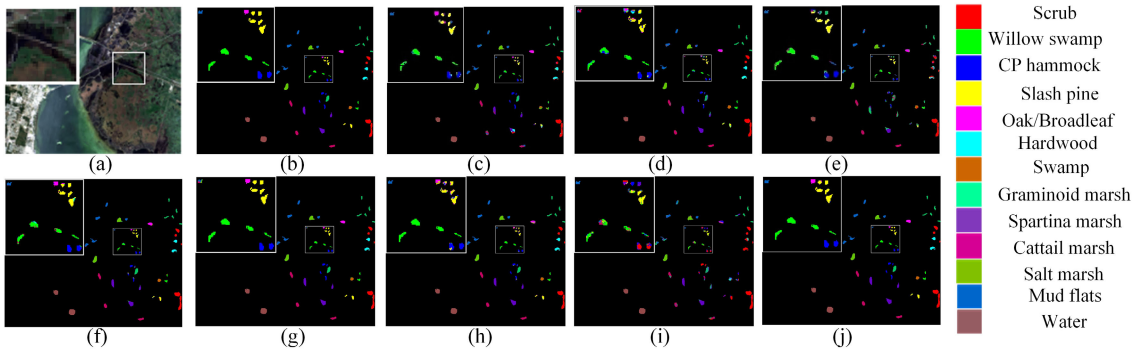


Fig. 9. Visualization maps of comparative methods on the KSC dataset: (a) false-color map, (b) ground truth map, (c) HybridSN, (d) MSDN, (e) S3EResBOF, (f) SSFRAN, (g) SSRN, (h) HSI-BERT, (i) SSFTT, and (j) LGGNet.

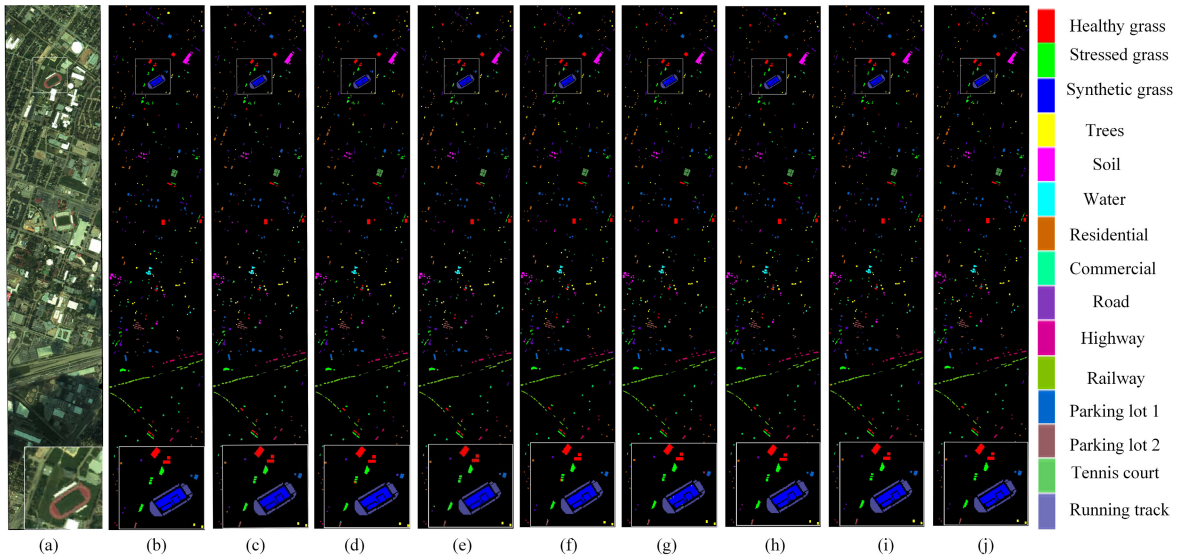


Fig. 10. Visualization maps of comparative methods on the HU dataset: (a) false-color map, (b) ground truth map, (c) HybridSN, (d) MSDN, (e) S3EResBOF, (f) SSFRAN, (g) SSRN, (h) HSI-BERT, (i) SSFTT, and (j) LGGNet.

learnable positional prompting and the local–global cross fusion network. The latter involves a sequence of operations encompassing 3-D CNN feature extraction, transformer feature extraction, and feature fusion. To prove the significant role of each component in LGGNet, ablation studies were carried out on the IP, KSC, HU, and PU datasets. This study involved the creation of seven distinct models, denoted as LGGNet_wC, LGNet_wC, LGGNet_wT, LGNet_wT, LGGNet_wF, LGNet_wF, and LGNet.

Specifically, LGGNet_wC represents a model that omits 3-D CNN feature extraction while keeping other settings in line with LGGNet. Similarly, LGNet_wC is an instantiation of

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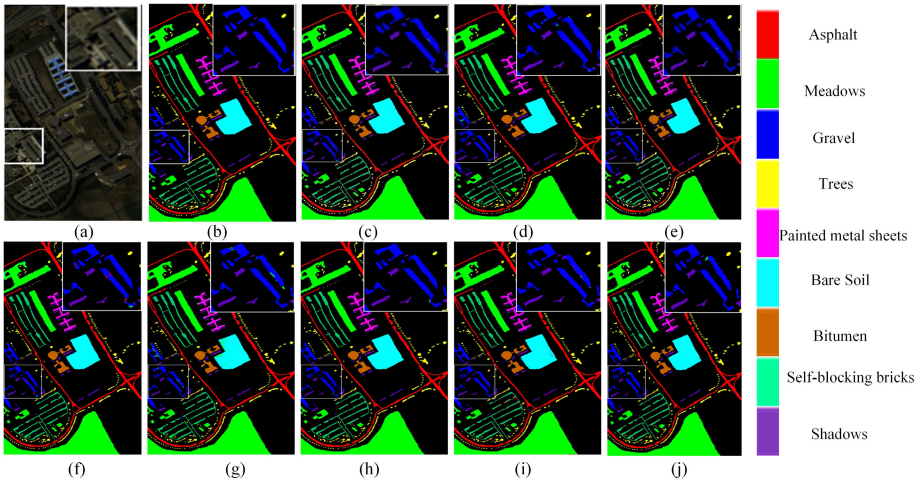


Fig. 11. Visualization maps of comparative methods on the PU dataset: (a) false-color map, (b) ground truth map, (c) HybridSN, (d) MSDN, (e) S3EResBOF, (f) SSFRAN, (g) SSRN, (h) HSI-BERT, (i) SSFTT, and (j) LGGNet.

TABLE VII

EXPERIMENT RESULTS OF LGGNET ABOUT THE ABLATION STUDY ON THE EXPERIMENTAL DATASETS

Name		LGNet_wC	LGGNet_wC	LGNet_wT	LGGNet_wT	LGNet_wF	LGGNet_wF	LGNet	LGGNet
Component	Gaussian-initialized Learnable Positional Prompts	×	✓	×	✓	×	✓	×	✓
	3D CNN feature Extraction	×	×	✓	✓	✓	✓	✓	✓
	Transformer feature Extraction	✓	✓	×	×	✓	✓	✓	✓
	Feature Fusion	✓	✓	✓	✓	×	×	✓	✓
IP dataset	Kappa(%)	94.52±0.38	**95.91±0.16**	86.27±0.49	**86.79±0.10**	98.04±0.03	**98.29±0.03**	98.44±0.02	**98.54±0.01**
	OA(%)	95.18±0.29	**96.41±0.12**	87.96±0.37	**88.40±0.07**	98.28±0.02	**98.50±0.02**	98.63±0.02	**98.72±0.01**
	AA(%)	96.05±0.46	**96.63±0.37**	84.10±1.57	**84.71±0.28**	96.47±1.03	**96.82±1.14**	98.28±0.53	**98.85±0.05**
	FLOPs(MMac)	0.20±0.00	0.22±0.00	4.66±0.00	4.90±0.00	5.28±0.00	5.28±0.00	5.31±0.00	5.56±0.00
	Param(K)	2.31±0.00	2.74±0.00	46.97±0.00	47.14	86.34±0.00	88.88±0.00	86.34±0.00	86.51±0.00
KSC dataset	Kappa(%)	97.66±0.97	**97.69±1.85**	92.99±0.34	**93.62±0.60**	99.24±0.02	**99.37±0.05**	99.61±0.01	**99.69±0.03**
	OA(%)	97.91±0.79	**97.93±1.51**	93.70±0.27	**94.27±0.49**	99.32±0.01	**99.43±0.04**	99.64±0.01	**99.72±0.02**
	AA(%)	96.28±1.69	**96.96±1.56**	90.95±0.91	**91.86±1.13**	98.91±0.04	**99.01±0.12**	99.33±0.03	**99.46±0.05**
	FLOPs(MMac)	0.20±0.00	0.22±0.00	4.66±0.00	4.90±0.00	5.28±0.00	5.28±0.00	5.31±0.00	5.55±0.00
	Param(K)	2.26±0.00	2.69±0.00	46.68±0.00	46.85±0.00	86.05±0.00	88.59±0.00	86.05±0.00	86.22±0.00
HU dataset	Kappa(%)	96.56±0.15	**98.23±0.18**	96.57±0.08	**96.77±0.12**	99.16±0.01	**99.23±0.01**	99.20±0.01	**99.34±0.00**
	OA(%)	96.82±0.13	**98.37±0.15**	96.83±0.07	**97.02±0.10**	99.22±0.01	**99.29±0.01**	99.26±0.01	**99.39±0.00**
	AA(%)	97.25±0.12	**98.58±0.10**	96.61±0.11	**96.68±0.09**	99.34±0.01	**99.40±0.01**	99.34±0.00	**99.46±0.00**
	FLOPs(MMac)	0.20±0.00	0.22±0.00	4.66±0.00	4.90±0.00	5.28±0.00	5.28±0.00	5.31±0.00	5.56±0.00
	Param(K)	2.29±0.00	2.72±0.00	46.88±0.00	47.05±0.00	86.25±0.00	88.78±0.00	86.25±0.00	86.41±0.00
PU dataset	Kappa(%)	99.15±0.01	**99.74±0.02**	97.68±0.02	**97.80±0.03**	99.66±0.00	**99.71±0.00**	99.80±0.00	**99.82±0.00**
	OA(%)	99.36±0.01	**99.81±0.01**	98.25±0.01	**98.34±0.02**	99.75±0.00	**99.78±0.00**	99.84±0.00	**99.86±0.00**
	AA(%)	98.90±0.02	**99.68±0.08**	97.20±0.04	**97.46±0.04**	99.42±0.00	**99.48±0.01**	99.60±0.00	**99.62±0.00**
	FLOPs(MMac)	0.20±0.00	0.22±0.00	4.66±0.00	4.90±0.00	5.28±0.00	5.28±0.00	5.31±0.00	5.55±0.00
	Param(K)	2.19±0.00	2.62±0.00	46.30±0.00	46.46±0.00	85.66±0.00	88.2±0.00	85.66±0.00	85.83±0.00

LGGNet_wC without the application of Gaussian-initialized learnable positional prompting. LGGNet_wT excludes transformer-based feature extraction, and LGNet_wT mirrors LGGNet_wT without Gaussian-initialized learnable positional prompting. LGGNet_wF bypasses the feature fusion method proposed in this article, instead amalgamating the output features of transformer and 3-D CNN by pooling and addition. Also, LGNet_wF is the LGGNet_wF without Gaussian-initialized learnable positional prompting. LGNet is the local–global cross fusion network without Gaussian-initialized learnable positional prompting. The experiment result of the ablation study on four experiment datasets is shown in Table VII.

Notably, the classification accuracy of LGGNet_wC, LGGNet_wT, LGGNet_wF, and LGGNet surpasses that of their respective counterparts LGNet_wC, LGNet_wT, LGNet_wF, and LGNet, respectively, as indicated by the instances of bold font in the table. In addition, there is no significant difference in the amount of computation and parameters between the methods with and without Gaussian-initialized learnable positional prompting. This observation substantiates that the Gaussian-initialized learnable positional prompting can improve the HSI classification performance in a lightweight way by learning the fine-grained importance of spatial pixels within a 3-D sample.

Comparatively, the classification performance of LGGNet outperforms that of LGGNet_wC, which proves the importance of 3-D CNN feature extraction to obtain low-level local features. Furthermore, the classification metrics of LGGNet eclipse those of LGGNet_wT, demonstrating the significant

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TABLE VIII
COMPARISON RESULTS BETWEEN LGNET_SA AND LGGNET ON THE EXPERIMENTAL DATASETS

Name	Dataset	IP dataset		KSC dataset		HU dataset		PU dataset	
	Model	LGNet_SA	LGGNet	LGNet_SA	LGGNet	LGNet_SA	LGGNet	LGNet_SA	LGGNet
Component	Spatial Attention Mechanism	✓	×	✓	×	✓	×	✓	×
	Gaussian-initialized Learnable Positional Prompts	×	✓	×	✓	×	✓	×	✓
Evaluation Metrics	Kappa(%)	98.46±0.02	**98.54±0.01**	99.55±0.06	**99.69±0.03**	99.30±0.01	**99.34±0.00**	99.79±0.00	**99.82±0.00**
	OA(%)	98.65±0.01	**98.72±0.01**	99.59±0.05	**99.72±0.02**	99.35±0.01	**99.39±0.00**	99.84±0.00	**99.86±0.00**
	AA(%)	98.59±0.11	**98.85±0.05**	99.29±0.14	**99.46±0.05**	99.46±0.01	**99.46±0.00**	99.58±0.01	**99.62±0.00**

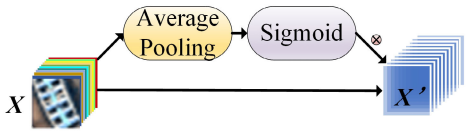


Fig. 12. Structure of the spatial attention mechanism.

role of transformer feature extraction in acquiring high-level global features. In addition, the superior classification performance of LGGNet compared to LGGNet_wF validates the efficacy of the proposed feature fusion method, effectively amalgamating disparate feature types.

2) Analysis of the Gaussian-Initialized Learned Positional Prompting: Gaussian-initialized learnable positional prompting makes the model pay attention to useful information embedded within the 3-D cube of HSIs by introducing learnable prompts. Similarly, the spatial attention mechanism achieves efficient information assimilation by delicately balancing attention weights across varying pixels within the input hyperspectral cube samples. Consequently, to conduct a comparative analysis of these two mechanisms, corresponding experiments were undertaken in the study. Specifically, this article presents LGNet_SA, which replaces LGGNet's Gaussian-initialized learnable positional prompting with the spatial attention mechanism while maintaining the same structural elements as LGGNet. The structure of the spatial attention mechanism is shown in Fig. 12. It primarily consists of an average pooling layer and a sigmoid function. The average pooling operates along the channel dimension to highlight significant regions. Also, the sigmoid function normalizes features to weight values between 0 and 1. These weights are then multiplied with the input, intensifying the focus on valuable spatial information within the sample of HSIs.

The comparison results between LGNet_SA and LGGNet are presented in Table VIII. It is evident that across all four experimental datasets, LGGNet consistently outperforms LGNet_SA, thus substantiating the superiority of the proposed approach. It is evident from the results across all four experimental datasets that LGGNet consistently outperforms LGNet_SA. This substantiates that in comparison between Gaussian-initialized learnable positional prompting and spatial attention mechanism, the former better captures the nuanced significance of spatial pixels. This phenomenon arises from the fact that spatial attention mechanisms commonly employ fixed-scale pooling or convolutional layers, which constrains the learning of parameters. In contrast, Gaussian-initialized learnable positional prompting dynamically updates prompts, relaxing these constraints and thereby fostering the acquisition of finer-grained features. This observation underscores the advantage of the proposed Gaussian-initialized learnable positional prompting.

Furthermore, to gain insights into the prompts learned through Gaussian-initialized learnable positional prompting, we visualize prompts obtained at different stages, as shown in Fig. 13. For each dataset, we recorded the weights of visual prompts at intervals of 1000 iterations within the range of 0–5000 iterations. It can be observed across the four experimental datasets that as the number of iteration steps increases, Gaussian-initialized learnable positional prompting undergoes autonomous parameter updates through backpropagation. This iterative process facilitates the acquisition of a nuanced understanding of the significance of fine-grained spatial pixel positions, enabling the prompts to adapt for the requirements of diverse HSI datasets.

3) Analysis of the Patch Size: The spectral–spatial classification method based on deep learning commonly employs the 3-D cube $X \in \mathbb{R}^{P\times P\times M}$ to obtain the latent information of HSIs. Here, the patch size represents the size $P \times P$ of the 3-D cube. It is worth noting that the selection of appropriate patch size is of great importance. If the patch size is too small, it will lead to inadequate receptive field of convolution and ineffective utilization of HSI features. Conversely, an excessively large patch size can escalate computational demands and even compromise model performance. To identify the ideal patch size for LGGNet across diverse datasets, a series of comparative experiments were conducted by varying the patch size between 9×9, 11×11, 13×13, 15×15, 17×17, and 19×19. The results on different datasets are shown in Fig. 14. When the patch size diminishes below 13×13, the OA of LGGNet shows a trend of gradual improvement. Notably, when the patch size reaches 13×13, LGGNet attains its peak classification performance in terms of OA and FLOPs, signifying an optimal point. Subsequently, when the patch size exceeds 13×13, the classification capability of LGGNet begins to decline. Therefore, 13×13 emerges as the preferred choice for LGGNet, striking a balance between classification accuracy and computational efficiency.

4) Analysis of the Band Number: The number of bands used in HSI classification methods affects the classification performance. An insufficient number of bands may compromise the effective extraction of spectral features from the HSI

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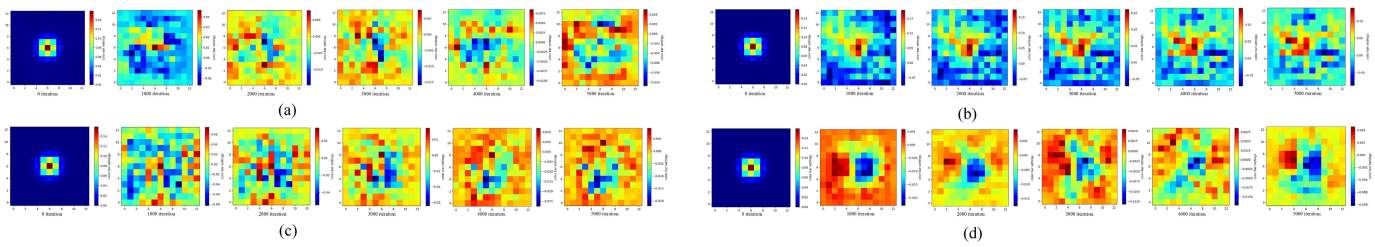


Fig. 13. Visualized prompts obtained at different stages on the experimental datasets. (a)–(d) Visualized prompts on the dataset of IP, KSC, HU, and PU, respectively.

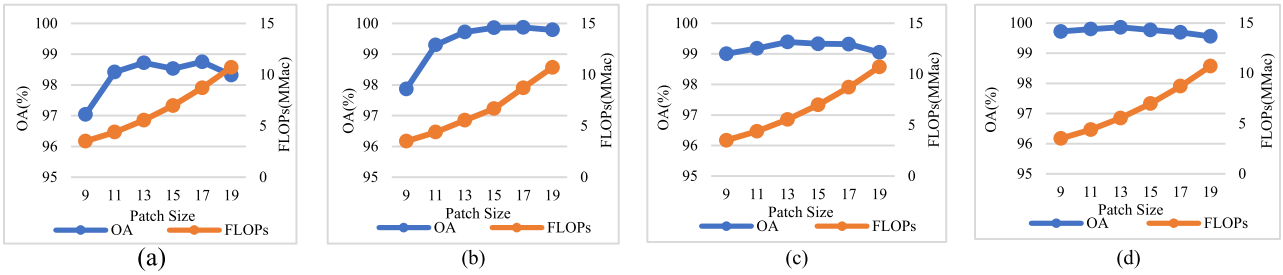


Fig. 14. Experiment results of LGGNet about the patch size on the experimental datasets. (a)–(d) Results on the IP, KSC, HU, and PU datasets, respectively.

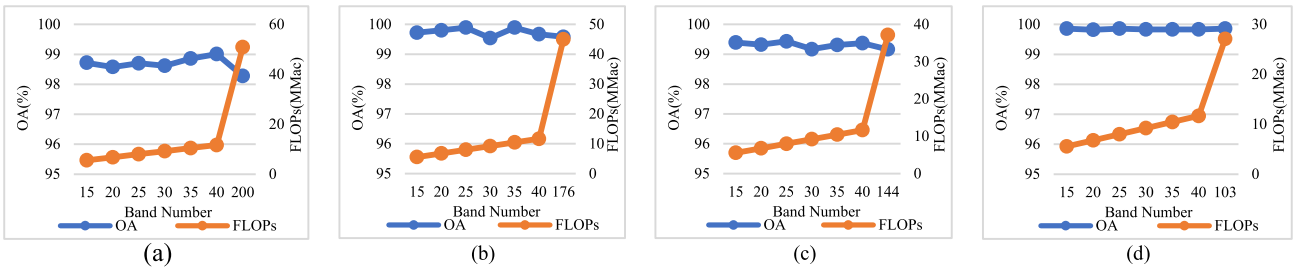
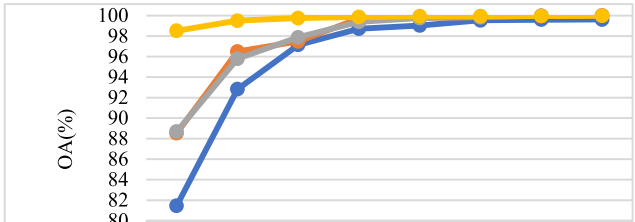


Fig. 15. Experiment results of LGGNet about the band number on the experimental datasets. (a)–(d) Results on the IP, KSC, HU, and PU datasets, respectively.

data, whereas an excessive number of bands might trigger the curse of dimensionality. To explore the influence of band number on the classification performance of LGGNet, relevant experiments were carried out. The results of these experiments are shown in Fig. 15. The experiments can be roughly divided into the method with PCA and the method without PCA. In the method with PCA, experiments with the band number ranged from 15 to 40 with the interval of 5 were implemented. Conversely, in the method without PCA, the band number adopted in the experiment is consistent with the spectral band number of the experimental datasets. On four experimental datasets, it is noteworthy that an increased band number does not necessarily correlate with heightened classification accuracy, yet it consistently escalates computational load. Through the comprehensive comparison of classification accuracy and computational complexity, it can be concluded that when the band number is 15, the performance of LGGNet is optimal. Thus, 15 was identified as the optimal band number for LGGNet across the four experimental datasets.

5) Analysis of the Training Sample Number: To scrutinize the sensitivity of LGGNet to the training sample number, relevant experiments were conducted. On four experimental datasets, the experimental results of LGGNet with training sample ratios of 1%, 3%, 5%, 10%, 15%, 20%, 25%, and 30% were tested, as shown in Fig. 16. A discernible pattern



Ratios	1%	3%	5%	10%	15%	20%	25%	30%
IP	81.46	92.82	97.14	98.72	99.03	99.53	99.6	99.62
KSC	88.53	96.5	97.48	99.72	99.8	99.85	99.97	100
HU	88.69	95.8	97.88	99.39	99.76	99.85	99.97	100
PU	98.52	99.5	99.76	99.86	99.93	99.94	99.95	99.98

Fig. 16. Experiment results of LGGNet about the training sample number on four experimental datasets.

emerges from the analyses of these experiments. Notably, when the training sample ratio is set at a mere 1%, LGGNet struggles to effectively capture and learn the underlying data information, subsequently leading to a subpar classification performance. As the training sample number progressively increases, there is a general trend of steady enhancement in classification performance. Achieving noteworthy results, when the training sample ratio attains 10%, LGGNet garners evaluation indicators surpassing 98% across all four datasets. Impressively, when the training sample ratio reaches 15%,

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these evaluation indicators soar above 99% across the diverse datasets. To demonstrate that LGGNet can achieve satisfactory classification results with limited training samples, the training sample ratio is set as 10% in the comparison experiment. Moreover, it is worth noting that the alteration in the training sample count has a minimal impact on LGGNet's classification performance with regard to the PU dataset in comparison to the IP, KSC, and HU datasets. This divergence is attributed to the PU dataset's ample sample availability, which permits LGGNet to comprehensively capture the data's distinctive characteristics. This observation underlines the correlation between data volume and the efficacy of deep learning-based HSI classification. Specifically, larger datasets tend to yield superior classification outcomes.

V. Conclusion

This article introduces the novel framework called LGGNet, which effectively addresses the limitations of conventional HSI classification methods by combining the strengths of Gaussian-initialized learnable positional prompting and a local–global cross fusion network. LGGNet stands out by excelling in both spatial importance learning and comprehensive integration of 3-D CNN and transformer architectures, thus efficiently harnessing local and spatial information for improved classification accuracy. Extensive experiments on four benchmark datasets underscore LGGNet's capability to achieve the state-of-the-art performance, even with a scarcity of training samples, all while maintaining a lightweight computational complexity.

Specifically, in LGGNet, the Gaussian-initialized learnable positional prompting method is introduced to tackle the challenges posed by spectral–spatial HSI classification approaches. This method efficiently addresses the issue of insufficient attention to pertinent pixels within 3-D samples by employing Gaussian positional encoding, latent information embedding, and backpropagation parameter updating. In addition, the local–global cross fusion network is devised to capture both local and global information from HSIs. This network incorporates various components, including the initial 3-D feature extraction module and the 3-D spectral–spatial residual module for 3-D CNN-based feature extraction. Furthermore, the transformer module is introduced for feature extraction from a global perspective, while the feature interaction module and token cross fusion module are designed for effective feature fusion. These interconnected elements collectively contribute to LGGNet's exceptional classification efficacy, robust generalization capability, and efficient computational complexity.

While the LGGNet model demonstrates commendable performance in hyperspectral classification tasks, it is essential to acknowledge that certain challenges persist. Like many other classification techniques that rely on spectral–spatial features, LGGNet employs the common practice of random sample division for training and testing. However, this approach can inadvertently introduce sample overlap between these two sets, leading to overly optimistic classification results.

The core concern here is the potential leakage of information from the training set into the testing set due to this overlap. Such information leakage can inflate the model's apparent performance, as it has gained insights from the very data it is supposed to generalize to. This issue raises questions about the model's real-world applicability and its generalization capabilities.

To address this challenge, we will explore advanced cross-validation techniques that can alleviate the issue of sample overlap, ensuring a robust evaluation of HSI classification models. In the context of classification methods, we will consider leveraging contextual information and domain-specific knowledge in future research. This will involve investigating semisupervised and self-supervised classification paradigms, thereby fostering further development in the field of HSI classification.

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